Logarithmic laws

1

a $\log_{10} 20$
b $\log_{10} 6$
c $\log_{10}\left(\frac{1}{4}\right)$
d $\log_5 198$
e $\log_{10}\left(\frac{35}{3}\right)$
f $\log_7 2$
g $3\log_{10} 18$
h $3\log_{10} 3$
i $2\log_5 2$
j $5\log_{10} 18$
k $\log_4 448$
l $\log_3 (10xy)$

2

a 1
b 3
c 3
d 2
e -2
f 2

3

a $7\log_4 x$
b $5\log(x+6)$
c $-\log(3x+7)$
d $\frac{2}{5}\log x$
e $5\log_6 y$
f $\frac{1}{2}\log_6 w$
g $\frac{1}{2}$
h $\frac{5}{4}$
i a
j 10
k $\frac{3}{2}\log_5 x$
l $\frac{15}{4}$

4

a $\log_9 u + \log_9 v$
b $\log_5 9 - \log_5 7$
c $\log_2 5 + \log_2 x$
d $\log_{10} 2 - \log_{10} x$
e $\log m + \log m$
f $5(\log 3 + \log x)$
g $\log p + \log q - \log r$
h $-\log x - \log y$
i $-7(\log 5 + \log x)$
j $\log 5 + \frac{2}{3}\log x$
k $\frac{1}{3}(\log 14 + \log x)$
l $4\log c - \frac{1}{2}\log d$

5

a $7\log x$
b $\log\left(x^5y^3\right)$
c $\log\left(\frac{x^8}{y^{\frac{1}{3}}}\right)$
d $\log\left(\frac{x^8}{y}\right)$
e $-35\log_{10} 5$
f $9\log_{10} 2$
g 2
h 2

6 $\log 2 + \log u - \log 3 - \log v$

7 $5\log x$

8

a $3\log u + 5\log v$
b $\frac{1}{3}\log v - \frac{1}{2}\log u$

9

a 5 b 2 c 9 d -4

e 2 f 3 g 5 h 2

10 2.105

11

a 1.447 b 0.222 c 3.121 d -0.407

12

a 1.24 b 0.31 c 1.78 d -0.62

e 2.02

13

a 7.8 b 1.4 c 7.3 d -1.6

14 $\dfrac{\log_4 20}{\log_4 3}$

15

a $\dfrac{\log_{10} 16}{\log_{10} 4}$ b $\dfrac{\log_{10} 0.9}{\log_{10} 3}$ c $\dfrac{\log_{10} \sqrt{5}}{\log_{10} 3}$

16

a i $\dfrac{\log_{10} 21}{\log_{10} 8}$ ii 1.46

b i $\dfrac{\log_{10} \sqrt{5}}{\log_{10} 4}$ ii 0.58

c i $\dfrac{\log_{10} 3^3}{\log_{10} 2}$ ii 4.75

d i $\dfrac{\log_{10} 105}{\log_{10} \pi}$ ii 4.07

17

a $p+q$ b $p-q$ c $2q$ d $2p+1$